TEAM 39 · GOKULA · VERIFIED-AGENT BADGE

UNCLONE

Verified-Agent Badge

customer-side agent verifies the bank's agent on a live phone call

Try it live — https://unclone-verified-agent.onrender.com

Source: https://github.com/TODO/VERIFIED-AGENT-BADGE/

UNCLONE Track 3 · Day 1 + Day 2 (combined)

Verified-Agent Badge

customer-side agent verification

UNCLONE Track 3 · Day 1

SLIDE 1

Problem

Who: a retail bank customer. Phone rings: “This is your bank's verification desk about a transaction of ₹86,499 on your card, above your limit.”

The moment someone tries to fool them: the “bank agent” on the line is an imposter — a live-cloned voice that sounds exactly like the real bank assistant, walking the customer into approving the payment or reading an OTP.

Why now: voice AI makes convincing real-time clones cheap; caller ID is spoofable, “the bank never asks for X” advice fails when the caller sounds like the bank. The customer has **no way to verify who is speaking**.

Stakes: one approval on a high-value/suspicious transaction = money gone, and the customer did everything “right”.

Judge link: https://unclone-verified-agent.onrender.com

SLIDE 2

Solution

Verified-Agent Badge — the user side verifies the bank's agent. (Bank's own agent already authenticates us; nobody authenticates it.)

The customer's side runs an AI agent that checks **every spoken turn** of the call against the enrolled, known voice of the genuine bank assistant.

The badge drives the decision: green = conversation may proceed to the transaction step; red = transaction **held**, customer warned, call escalated.

Two scenarios, both live in the prototype: **Call 1** above-limit (genuine agent → PROCEEDS), **Call 2** suspicious (imposter hijack → HELD).

Zero customer effort: verification happens automatically on their side of the line.

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SLIDE 3

DetectifAI (tools disclosed here per rules 5.2)

Both APIs, one call: DetectifAI POST /v1/speakers/verify with detect_deepfake=true → speaker verification (matches the caller against enrolled bank_agent_1) and deepfake detection (is this speech AI-generated?) per turn.

Where it sits: middle of the call flow — browser call screen → FastAPI backend → DetectifAI → badge verdict → transaction decision. Taken out of the product, the badge has nothing to stand on and the product stops existing.

The decision it drives: match + genuine → approve (PROCEEDS); no_match or synthetic → block (transaction held, warn customer).

Live results from the prototype today: genuine voice → match, similarity 0.987 (thr 0.53), deepfake real 0.008 (thr 0.62); attacker TTS → no_match, 0.151, deepfake synthetic 0.999, per-4s synthetic segments flagged.

Tools used: DetectifAI (verify + deepfake detection) · FastAPI/Python · edge-tts (team-generated attacker voice) · ffmpeg · Render · GitHub · built with an AI coding assistant (Codebuff/Claude Code).

Judge link: https://unclone-verified-agent.onrender.com

SLIDE 4

Impact

Who it protects: every customer on an outbound bank/NBFC/insurer verification call — the exact fraud moment DetectifAI itself was founded on.

What changes: the imposter call ends **before** the customer speaks a single secret; the genuine call gets a visible trust signal instead of anxiety.

Measured on the working prototype: ~2.5 s end-to-end per spoken turn; both decisions demonstrated live (approve 0.987 vs hold on 0.999-synthetic clone); no hardware, no customer onboarding beyond voice enrollment (15–60 s).

False-alarm posture: red never locks the customer out — it steps up to an alternate verified channel (in-app confirmation), so friction is bounded.

Judge link: https://unclone-verified-agent.onrender.com

SLIDE 5

Scale (incl. Day 2 plan)

Day 2 plan (nine hours, judge-gap driven):

1. **Streaming deepfake check —** feed the live call's audio continuously (DetectifAI's segment scoring already visualised in the UI) instead of per turn; deeper use of the API.

2. **Verified-badge handshake —** extend to agent-to-agent: any voice agent presents/enrolls its voice; a protocol-level “verified agent” attestation both directions.

3. **Phone channel path —** bridge the same check to PSTN/WhatsApp-audio calls (audio in → same API → same badge).

Scale: per-turn check costs 50+50 credits and ~2.5 s today — cheap enough to run on every suspicious transaction; enrollment reuses banks' existing voice-consent flows; the badge client is one API wrapper any call surface can embed.

Demo link (no login): **https://unclone-verified-agent.onrender.com**

Judge link: https://unclone-verified-agent.onrender.com

SLIDE 1 · PROBLEM

The phone scam no amount of care can stop

Your phone rings. **This is your bank about a Rs 86,499 payment above your limit.** Name right, details right, voice right.

The voice is a **computer clone** of the bank's agent. You cannot tell — no amount of care on your side helps.

The old guardrails are dead: caller ID is faked, the caller knows everything, and **they don't need you to tell them anything** — they need you to agree.

Our idea: **let the customer verify the caller.** Nobody is on the customer's side of the call today. We are.

Try it live — https://unclone-verified-agent.onrender.com

SLIDE 2 · SOLUTION

A traffic light for the phone call

An app on **your** side of the call checks the audio it is given — the demo runs recorded call clips and your live mic through it, every 6 seconds: **is each slice the real enrolled agent, or a machine copying them?** (Capturing the phone line itself is the next version.)

GREEN · GO

Real agent voice, no machine parts. Carry on with the call.

AMBER · DOUBLE-CHECK

Can't be sure. Confirm the payment in the app instead.

RED · STOP

A fake was caught. The call is cut, the payment blocked.

The red light is not a warning — **watch the demo: the audio stops mid-sentence** and the transaction flips to payment blocked.

And it works on voices it has never heard: our judges can enrol their own voice and try it live.

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SLIDE 3 · THE ENGINE

Every attack — and which rule catches it

Four checks per 6-second slice: **two AI-model questions** (identity, machine-voice) plus **two app-side checks** (room-tone forensics, replay). Each targets a different failure mode.

#	WHAT THE ATTACKER DOES	WHAT THE PIPELINE SEES	WHICH RULE CATCHES IT	RESULT
A	**Clone of a DIFFERENT person's voice (cheap TTS — our Call 2)**	similarity 0.13 « 0.53	Voice-fingerprint gate	**HOLD**
B	**Same-agent-timbre clone that fools identity (0.77 — our Call 3)**	Identity PASSES, average says "real" but one 4-s segment scores 0.99 synthetic	Hot-segment rule (any slice ≥ 0.90)	**HOLD (worst segment, not average)**
C	**Clone that also fools the deepfake (newer engine, unseen in training)**	Deepfake score low, similarity high	Forensics clue (app-side): a digital render has no room tone → artifact ≥ 0.85 (PCM WAV only)	**STEP-UP (never auto-approve)**
D	**Replay of the agent's REAL recorded audio**	Everything passes — nothing new to detect	Content fingerprint: identical audio re-sent	**HOLD**

Layer principle: each check targets a failure mode the others miss — identity catches "wrong voice"; the segment check catches "right voice, machine words"; forensics adds a best-effort room-tone backstop for new unseen engines; replay catches "it's just the recording".

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SLIDE 4 · IMPACT

What one caught call is worth

Median loss per imposter scam: $800 (FTC 2024 Data Book). Bank-wire scams skew higher — this is the conservative anchor.

Cost to check a risky call: about 45 cents — spent only on flagged calls, never on all traffic.

At one big bank scale: about $9.3 M/yr of prevented losses. That's the five-frame model: 1.56 M risky calls x 1.33 % scams x $800 x 80 % caught x 70 % stopped.

False alarms are cheap: worst case, the customer confirms in the app. Nobody is ever locked out.

~13x

return on the checking cost: about $700 K/yr of checks prevents ~$9.3 M of scams — roughly $42 per fraud caught, $60 per prevented, against an $800 median loss.

Latency: p50 ~ 0.6 s per slice (p95 ~ 2.6 s) on the deployed build — inside the decision window.

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SLIDE 5 · SCALE

Banks pay for it — and starting needs zero data

Who pays: the bank's fraud team. A bank can't stop a **perfect clone** from inside its own call centre — the clone is its own agent's voice. Only the **customer's side** can.

Starting takes a day, not a dataset: banks already record their agents for compliance. Take **15 to 60 s of that recording**, enrol it — done. **We train nothing.**

The honest scoreboard: live now — 3 attacked calls + your own voice. Attacks are **held** (red, call cut); errors and missing evidence fail safe (amber step-up) — neither ever auto-approves. Next — the phone-network version. We say which is which.

First pilot: 6 weeks, flagged calls only. Success = one caught scam with the full evidence log shown — then scale.

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SLIDE · SOURCE & THANKS

Project: UNCLONE — Verified-Agent Badge

Team: Team 39 — Gokula

What it does: customer-side agent verifies the bank's agent on a live phone call, every 6 seconds, before any money moves

Try it live: https://unclone-verified-agent.onrender.com

Source code: https://github.com/TODO/VERIFIED-AGENT-BADGE/

Built with: DetectifAI (speaker verification + deepfake detection), FastAPI, Python, edge-tts (team-generated attacker voice), ffmpeg, Render, GitHub

Thank you — questions welcome